LI-7200RS CO_2/H_2O gas analyzer maintenance

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables. Disassembling the instrument or detaching or attaching head cables while the instrument is powered on may damage the instrument or result in data loss.

Before performing maintenance to the sensor, please check thoroughly the details of that procedure in this document, if any questions arise do not hesitate to contact David Trejo (trejo_cancino.david_ulises@uqam.ca).

After conducting any maintenance protocol, register it in EcoFlux field notes to have a follow-up of each sensor and station. If something is wrong with any equipment, please contact David Trejo Cancino (trejo_cancino.david_ulises@uqam.ca) to register it in the database and proceed with the required diagnostics and repairs.

Summary

Procedure	Frequency	Items required
Replacing the internal chemicals	Once a year or every 6 months in a hot humid climate.	• Desiccant bottles. • Flat-head screwdriver.
Cleaning the gas analyzer optical path	Highly dependent on the air quality in the environment. In clean-air environments, the frequency of cleaning may be as low as several months. In environments or periods with high amounts of salts, pollen, or dust, the frequency may be as high as every one to two weeks.	• Demineralized water. • Microfiber soft cloth. • Mild soap. • Isopropyl alcohol.
Replacing the thermocouples	Replace if the wire is broken or there are wrong readings.	• 7/64” hex key. • Thermocouple.
Replacing the LI-7550 fuses (2)	If fuses are blown. Check resistance, if should be < 1 Ohm.	• Flat-blade screwdriver. • Power supply fuse. • 2A Nano2 SMF Fuse. • Ohmmeter
Replacing SmartFlux 2 fuses	If the SmartFlux fails to power up. If any fuse is blown. Measure with an ohmmeter, if one of them is an open circuit, replace it.	• 3A fuses. • 1A fuses. • Flat-blade screwdriver. • Ohmmeter.

Cleaning the intake cap and screen	Clean before the flow drive exceeds 90%	• Ultrasonic bath • Compressed air • Boiling water
Cleaning the 2-micron intake filter	Clean before the flow drive exceeds 90%	• Ultrasonic bath • Compressed air • Boiling water
Flow module maintenance	If the fuse is blown.	
Software Updates	When new software is released	
User Calibration: Zero and Span	Check in monthly intervals	

Replacing the internal chemicals

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

There are three small plastic bottles, each containing Ascarite II and magnesium perchlorate, in the upper and lower analyzer housings. The chemicals keep the source and detector housings free of CO2 and water vapor. The bottles should be recharged with fresh chemicals annually (or every 6 months in hot, humid climates). Replacement "charged" bottles are available from LI-COR in sets of three under part number 7200-950.

Note: Calibration shifts will occur if CO2 and H2O are not kept out of the analyzer housing.

To change the sensor head soda lime and desiccant bottles:

1) **Remove the chemical bottles**
   The plastic bottles are in the lower analyzer housing in the sensor head. Remove the mounting bracket from the analyzer. Then remove the thumbscrew and thread it into a cap. Pull straight out to remove the plug.



Figure 1: Remove the thumbscrew, thread it into the bottle covers, and then pull straight

2) **Fill the bottles half full of Ascarite II first, followed by the magnesium perchlorate.**
3) **Place the Teflon membrane in the lid to keep the chemicals from spilling into the detector housing.**
4) **Insert the recharged bottles into the analyzer housing cap first. Replace the bottle covers.**

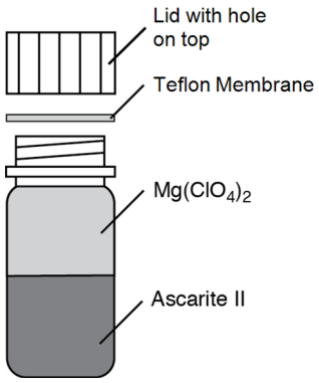




Figure 2: Insert the recharged bottles cap first. Make sure the caps have holes in the top.

5) **Use the retention screw to remove the bottle cover on the upper sensor head housing and insert the new, recharged bottle, cap first.**

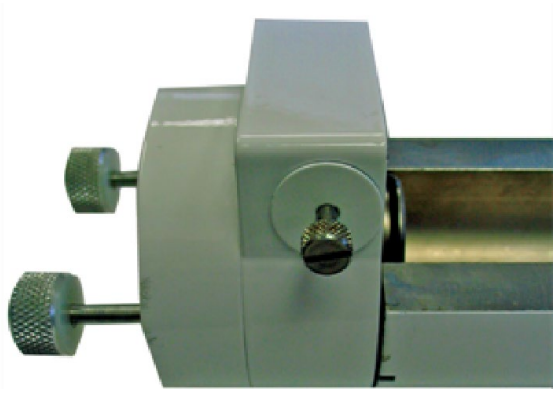


Figure 3: Remove the upper sensor head housing bottle cover and replace the chemical bottle. Insert the bottle cap first.

6) **Replace the cover retention screw and reattach the mounting bracket.**
   After installing, allow at least 24 hours (with the instrument powered on) to scrub the housing; otherwise, the instrument may still drift.
7) **Set the CO2 and H2O zeros. Check the zero again, if possible, after one or two days.**

Cleaning the gas analyzer optical path

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

The LI-7200RS windows should be kept clean through regular maintenance. The analyzer is specifically designed to make it easy to remove and clean the cell without the use of tools.

When operating without an intake filter, the cleaning frequency is highly dependent upon the air quality in the environment. In clean-air environments, the frequency of cleaning may be as low as several months. In environments or periods with high amounts of salts, pollen, or dust, the frequency may be as high as every one to two weeks.

To open the optical bench, loosen the two knurled knobs on the top of the LI-7200RS sensor head, and then pull the top of the sensor head out away from the optical bench.



Figure 4: Slide the top end of the sensor head away from the bottom to remove the optical path.

The sapphire windows are extremely durable and resistant to scratches. Clean them with a mild detergent or glass cleaner and a soft, lint-free cloth.

The optical bench has a PVC insert as the optical path. You can use mild soap and water, isopropyl alcohol, vinegar, or water to clean the optical path. **Do not use acetone, ammonia, chlorine, or wire brushes to clean the path, as irreparable damage to the PVC insert can occur**. Cleaning swabs in the spare parts kit (part number 610-05315) are to clean the optical path.

Reassemble the sensor head and perform a zero and span calibration.

Replacing the thermocouples

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

The fine wire thermocouple should be replaced if the wire is broken or if it reads incorrectly. There are fine wire temperature thermocouples located in the air inlet and outlet ports that measure the air temperature of incoming and outgoing air. Do not insert ¼" Bev-a-line or other small diameter tubing into the air IN or air OUT fittings, as you can break the fine-wire thermocouples present in each of the fittings. If a thermocouple does break, there are two replacement thermocouples in the spare parts kit (part number 9972-007). Follow these steps to replace either air temperature thermocouple:

1) **Loosen the two knurled knobs on the top of the LI-7200RS sensor head and remove the optical path.**
2) **Use a 7/64" hex key (part number 610-03745) to remove the eight hex screws on the back of the optical path.**
   It should slide out easily.



Figure 5: Hex screws positions

3) **Separate the optical path from the inlet/outlet assembly.**
   Be careful not to strain the tubing that connects the two.
4) **Remove the small thermocouple circuit boards.**
   The entire assembly, including the circuit board, is replaced as a unit. There are two pins on the underside of the circuit board. Lift up on the end of the board to remove the thermocouple assembly and discard. Make sure the two O-rings are present on the new board (one is tacked onto the thermocouple board; the other sits between the board and the opening), and insert the new assembly, making sure the two pins are inserted into the connector.

Reassemble the optical bench.

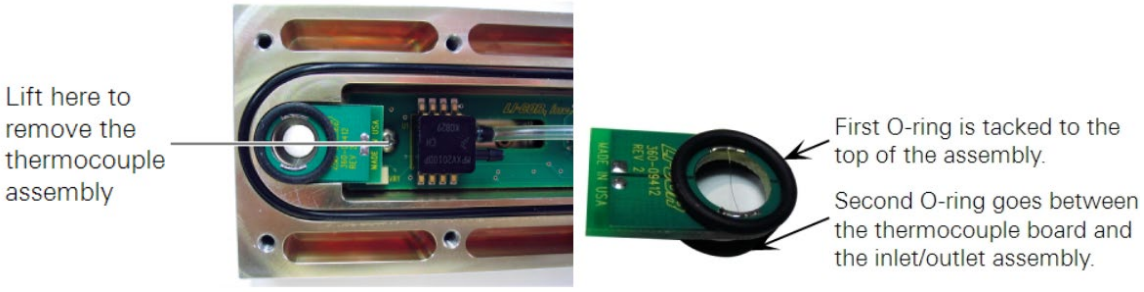


Figure 2: Remove the old thermocouple assembly. The replacement is part number 9972-007. An O-ring goes on each side of the thermocouple assembly. One is tacked to the board. The other is loose; it goes on the opposite side of the board as the glued one.

5) **Reassemble the analyzer and perform a leak test to ensure that the o-rings are seated properly.**

Follow these steps to perform the leak test:

- **Attach the sensor head to the Analyzer Control Unit.**
- **Remove the tubing from the pressure inlet on the sensor head. It is attached to the small hose barb between the Air IN and Air OUT ports.**
- **Cut a short length (6" or so) of 1/16" I.D. Tygon tubing.**
  There is a length of tubing in the spares kit (part number 222-09579). Attach this short length of tubing to the pressure inlet hose barb on the sensor head.
- **Connect the instrument to your computer with the Ethernet cable.**
  Open the LI-7200RS PC software and Connect to the instrument.
- **Display *Head Pressure (kPa)* in the software dashboard.**
  To change the variable displayed in the main data windows, click on the data value and select Head Pressure (kPa) from the list.
- **Gently blow into the length of the tubing connected to the pressure inlet.**
  You should see a negative pressure displayed at Head Pressure in the dashboard. Alternatively, if you inhale, you should see a positive pressure. You can also push or pull air through the tubing using a syringe.
- **Observe the Head Pressure reading over about 5 seconds.**
  If the reading changes by an integer or more during this time (e.g. from -10.625 to -9.625), the sensor head may have a leak. Disassemble the sensor head optical bench and make sure the o-rings are seated properly. Repeat the leak test until the reading is stable within one integer over 5 seconds.
- **Zero and span the instrument.**

Replacing the LI-7550 fuses

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

The LI-7550 has two fuses – one for the power supply and one for the accessory. To check a fuse, inspect it for evidence that it is burned, and use an ohm meter to check the resistance. Resistance of <1 Ω (ohm) indicates that the fuse is OK.

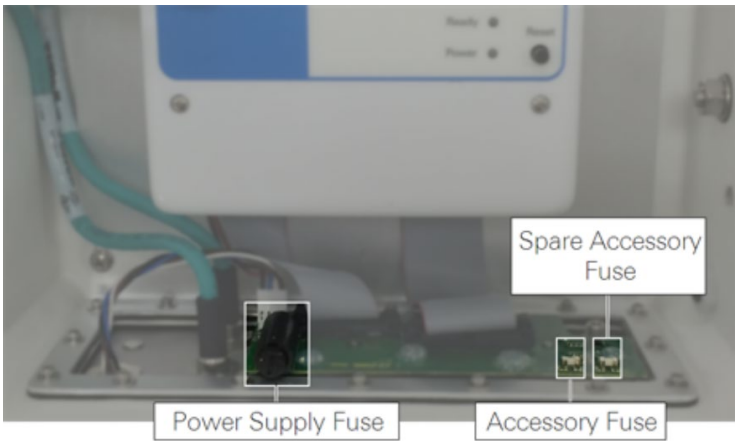


Figure 7: LI-7550 fuse locations.

Power Supply Fuse

The power supply is protected by a 5 A 125/250V, 5 × 20 mm fast-blow fuse (part number 439-04214). If the battery or other power source fails to power the instrument, check to see if the fuse has blown.

The fuse is in the lower left-hand corner, as shown in the figure.

Replacement fuses (part number 439-04214) are in the spares kit. Use a flat-blade screwdriver or your thumb to push down on the fuse holder cap and turn counterclockwise to release the cap.

Accessory Fuse

The accessory uses a 2A Nano^2 SMF Fuse. There is one spare fuse included in the LI-7550. If the heated intake tube will not power on or continuously triggers an error, check the fuse and replace it if necessary.

Replacing SmartFlux 2 System fuses

Important: Power off the SmartFlux System. Remove the power and data terminal strips. Take care to avoid damaging nearby electronic components when removing fuses. Especially avoid prying on top of other electronic components.

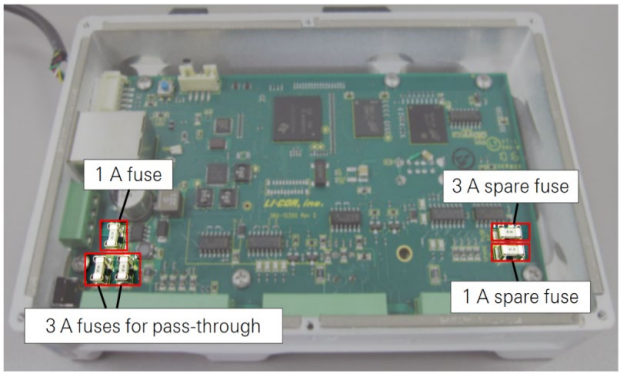


Figure 8: Fuses inside the SmartFlux 2 System

Power for the SmartFlux System and the pass-through power supply are protected by fuses. Pass-through power flows through the SmartFlux System to pins on ports 1, 2 and 3 to power a connected sonic anemometer. To change the fuses, power off the SmartFlux System. Remove the power and data terminal strips. Remove the six screws on the side of the case and remove the case lid.

3.0 Amp fuse for pass-through power

The SmartFlux System uses two 3.0 A fuses between the Power IN connector and the serial ports. One of these fuses is for the power wire to the anemometer or other connected device. The other 3.0 A fuse is for the power return wire and protects the system in case problems such as improper grounding cause a power surge. If the SmartFlux System boots up and has power, but the PowerOUT LED does not light up, it is likely that one of the two 3.0 A fuses has failed.

1) **Measure both 3.0 A fuses with an ohmmeter. If one of them is an open circuit, replace it with the spare 3.0 A fuse.**
2) **Remove the failed fuse and the spare 3.0 A fuse by gently prying under each fuse with a small flat-bladed screwdriver.**
3) **Gently press the good (spare) fuse into the fuse holder.**
   If both 3.0 A fuses are blown, you will need to order another 3.0 A fuse (part number 348-16020).
4) **Replace the lid and six screws on the SmartFlux 2 case.**

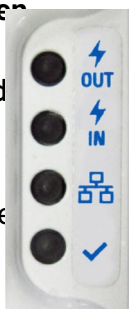


Figure 9: Lights of the Smart Flux 2

Important: A blown 3.0 A fuse usually indicates a wiring problem between the SmartFlux System and the sonic anemometer or other connected device. Check for proper system grounding and

remove possible causes of a short circuit in the power cable for the connected device. Also, while this fuse is rated at 3.0 A, keeping power use at 2.0 A or less will ensure a nearly unlimited lifetime for the fuse

1.0 Amp fuse

If the Power OUT LED lights up, but the Power IN LED does not light up, it is likely that the 1.0 A fuse has failed.

1) **Measure the 1.0 A fuse with an ohmmeter.**
   If it is an open circuit, replace it with the spare 1.0 A fuse (part number 348-13880).
2) **Remove the failed fuse and the spare 1.0 A fuse by gently prying under each fuse with a small flat-bladed screwdriver.**
3) **Gently press the good (spare) fuse into the fuse holder.**
4) **Replace the lid and six screws on the SmartFlux case.**

Important: A blown 1.0 A fuse may indicate a problem with the SmartFlux 2 or 3 System. If the spare 1.0 A fuse also fails when installed, contact LI-COR support.

Cleaning the intake cap and screen

To clean the cap and screen, remove the intake cap assembly and back-flush it with compressed air or water. You can immerse the intake cap assembly in boiling water for a few minutes or soak it in an ultrasonic water bath. Usually, you should not have to remove the screen from the cap. If you do need to remove the screen:

1) **Insert a small screwdriver between the screen and the intake cap.**
   Use caution to avoid damaging the cap and deforming the screen.
2) **Carefully pull the screen out of the cap.**
3) **Assembly is the reverse of removal. Be sure the screen is seated fully in the cap.**

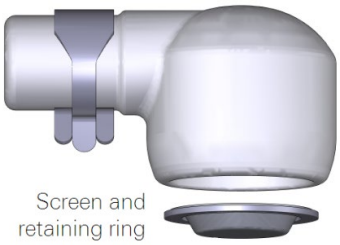


Figure 10: Intake cap and screen.

Cleaning the 2-micron intake filter



In the Carbonique sites, we have exchanged the default 2-micron intake filter for a 15-micron one. This allows the system to have good filtering while not relying so much on maintenance.

How do you know when to clean the filter? Dust will accumulate in the filter over time. The rate of dust accumulation will depend upon the amount of dust in the air and the particle size of dust. Therefore, it is difficult to predict exactly when the filter should be cleaned.

Figure 11: Default 2-micron intake filter

As dust builds up in the filter, the flow module will use more power to maintain the flow rate. So, the best indicator of the filter status is a change in the power required to run the flow module. When you install a new filter, record the Flow Drive (%), or log that variable with your data. Over time, compare the flow drive % with the initial value. The % will increase as the filter becomes clogged, indicating that the pump must work harder to move air through the filter.

Clean the filter before the Flow Drive (%) exceeds 90%.

Cleaning: If possible, use an ultrasonic water bath, then back-flush the filter with compressed air and allow the filter to dry before reinstalling. Otherwise, immerse the filter in boiling water for a few minutes or soak it overnight and then back-flush it with compressed air. As a good practice, clean the intake cap whenever you clean the filter.

The ferrules do not need to be removed from the intake tubes when the filter is removed. They can be reused.

Flow module maintenance

The flow module requires little routine maintenance.

Changing the fuse

The fuse in the flow module case can blow if power is applied with reverse polarity. Open the instrument case and unscrew the fuse cover to replace the fuse. The replacement fuse is a 5 A fast-blow, 125 V, 5x20 mm type, available from LI-COR under part number 439-04214.



Figure 12: Unscrew the fuse cover and replace the fuse.

Updating the software

The latest software is available from licor.com/7200rs-software. Update all components in the system before deploying it.

We recommend running the most current software at all times, including both the embedded and interface software for the LI-7500A/RS or LI-7200/RS, EddyPro, and embedded software on the SmartFlux System. Check the installed version of the embedded analyzer software by clicking the Diagnostics tab.

On the support website, select your instrument, then select *Software*. Download both the *Instrument (Embedded) Software* and the *Windows Interface Software*.

IMPORTANT: Read this before updating!

Firmware version 8.9 requires USB drives to be installed in the SmartFlux 2 System and the LI-7550 for dependable performance. Be sure to install USB drives in both the LI-7550 and the SmartFlux 2 System before installing the update. Also:

- Clear data from both USB drives prior to initiating the update. This ensures that there is adequate memory available to perform the update.
- Reapply the FluxSuite settings after the update is complete.
- Reapply data repository settings after the update is complete.
- To optimize performance of the LI-7550 with the SmartFlux 2 System, set RS-232 output to 0 Hz.

Embedded instrument software

To update the embedded instrument software:

1. **Make sure your gas analyzer sensor head is connected to the LI-7550 Analyzer Interface Unit. Power up the gas analyzer. Connect your computer to the system Ethernet switch or to an open Ethernet port inside the LI-7550 using an Ethernet cable with standard RJ45 connectors.**

Important: There must be a USB drive in the LI-7550 USB port. For the best speed and performance, it's best to use a USB drive with no data on it.

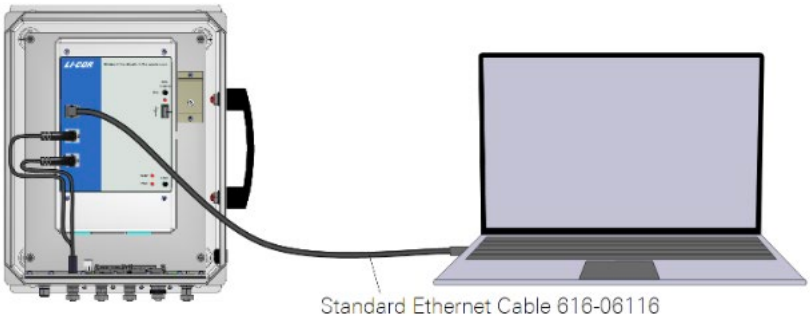
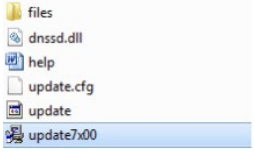


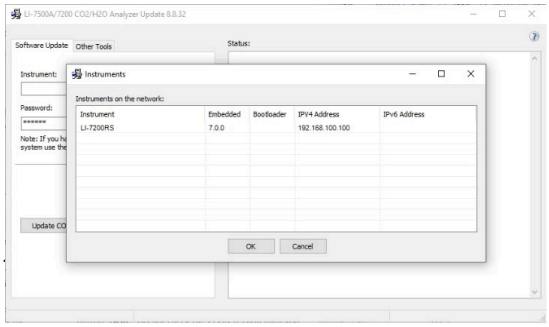
Figure 13: Computer connected to system Ethernet switch.

The Instrument (Embedded) Software is a zipped file named something similar to *LI-7xxx_embedded-7.0***.** Unzip the files and save them.



2. **Double-click the file called *update7x00*.**

3. **Click *Browse*... and select the instrument from the list. You may need to allow the application to pass through your computer's firewall.**



4. **Select the gas analyzer from the list and click *Update Software*.**
   The update will take about 5 minutes.

Important: Do not close the software, let your PC go to sleep, or power off the instrument during the update process. The software will notify you when the update is complete.
If the update fails for any reason, repeat steps 2 and 3.

If you are updating from embedded version 4.x.x or earlier, you must also run the FPGA update. To check if this update is necessary click the *Check FPGA* button.
Note: Do not check the FPGA version until you update the system software.

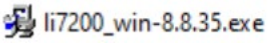
If the FPGA update is needed the *Update FPGA* button will become enabled in the application. The FPGA update takes approximately 10-12 minutes.

Important: Do not close the software or reset the instrument while the update is taking place. The software will notify you once the FPGA programming is finished.

Troubleshooting: Getting an update error? There must be a USB drive installed in the LI-7550 USB port before you can update. Also, if the drive is inserted for the first time, the instrument will take a few minutes to connect with the drive.

Windows interface software update

To update the Windows interface software for gas analyzers:

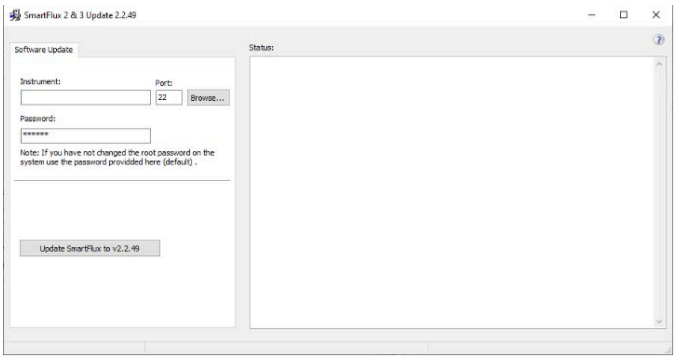
1. **The Windows interface software is an executable file named something similar to *LI-7x00_win8.0*. Double click to launch the installer.**
   

2. **Follow the Windows Installation wizard to install the application.**

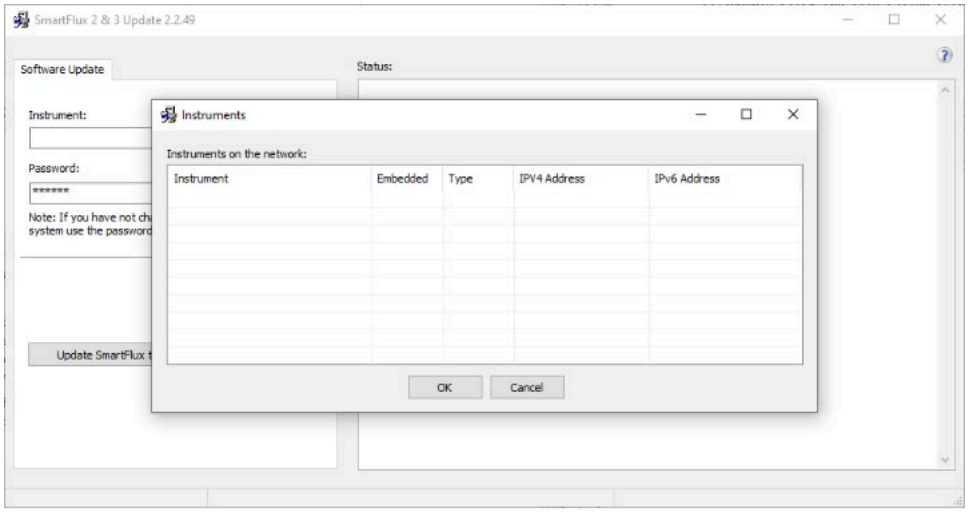
SmartFlux 2 System software update

To update the software:

1. **Save the SmartFlux software update package to your computer.**
   It is a collection of files in a zipped folder. Extract the contents of the file.
2. **Double click the file called *update7x00.exe***



3. **Click *Browse* to view systems. You can also enter SmartFlux IP address or hostname as the *Instrument*.**



You may need to allow the updater to pass through your computer firewall. Your computer firewall may prompt you if required. All SmartFlux 2 and 3 Systems connected to your network will be visible in the window.

4. **After selecting system or entering the IP address, click *Update Firmware*.**
The updater will load the new system files. This process takes about 30 seconds to a minute to complete the update. Do not power off the device while it is updating.

User calibration: zero and span

The LI-7200RS's measurement accuracy depends upon its calibration. There are two major components to the calibration: 1) determining the values of calibration coefficients, and 2) setting zero and span. During a factory calibration, both of these steps are performed. The values of the coefficients that are determined should be valid for several years. The zero and span adjustments are used to bring the LI7200RS's actual response into line with its previously determined factory response, at least at two points.

Important: Check the zero and span at regular intervals; monthly at first, and then adjust the frequency according to the stability of the instrument.

How stable are zero and span?

The analyzer's zero is primarily affected by temperature, and the state of the internal chemicals. The internal chemicals should be changed annually. The zero's response to temperature is relatively small (typically 0.1 or 0.2 ppm per °C for CO_2 or 0.01 mmol/mol/°C for H_2O). Also, this drift is measured at the factory, and subsequently compensated for in software, so the effective zero drift should be quite small. Therefore, the zero should be quite stable over a several month period, but you might want to check it after an extreme temperature change (>30 °C).

The analyzer's span is affected by temperature, pressure, and the state of the internal chemicals.

- Temperature: A 10 °C change will typically change the H_2O span by 1 to 2%. For CO_2 at ambient concentrations, the span is very insensitive to temperature.
- Pressure: A large pressure change (40 kPa) will affect the CO_2 and H_2O spans by <1%, for ambient CO_2 concentrations (>400 ppm) and high humidities (20 mmol/mol). So, diurnal pressure variations should not be a concern.
- Chemicals: Reduced internal chemical effectiveness will affect the span, but the effect on the zero will be much more pronounced. In summary, span stability is mostly a concern with H_2O, when there are large temperature changes.

Checking the zero

Flow a dry CO_2-free air through the optical path and check the analyzer reading in the software dashboard. A suitable source of air for setting the zero can be generated with chemical scrubbers (such as soda lime for removing CO_2 and magnesium perchlorate or Drierite® for removing water), or obtained from a cylinder of zero-grade gas.

When using chemical scrubbers, make sure that the chemicals are fresh and that air goes through the chemicals in the right order: Soda lime first, desiccant last (if the desiccant is Drierite®, allow time for the CO_2 to "wash out" of it). Use a small pump to push the air through the gas analyzer.

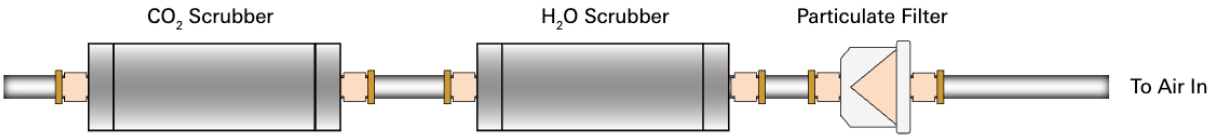


Figure 14: Order of scrubbers and flow of air

When using a cylinder, make sure that the air in the cylinder really is CO_2-free since a typical cylinder of standard grade nitrogen might have as much as 20 ppm of CO_2 in it. Compressed cylinders may be at pressures of several thousand pounds per square inch; before using them for calibration, they should be fitted with a regulator to reduce the pressure down to a range of around thirty pounds per square inch. Set the flow rate from 0.5 to 1.0 lpm.

Checking the span

Check the CO_2 span with a cylinder of CO_2 in air at a concentration that is at the higher end of the range of concentrations which may be encountered during measurements. For example, a 500 to 1000 PPM cylinder of CO_2 in air, which has been verified to be accurate to within at least 1% would be a suitable choice for many applications. Be cautious, as the stated value of the calibration cylinder may be significantly different from the actual gas concentration. Set the flow rate to 0.5 to 1.0 lpm. Check the analyzer reading in the software dashboard.

For the water vapor span, a convenient standard to use is a dew point generator such as the LI-610. To avoid condensation problems choose a dew point temperature that is about 3 to 5 °C below the ambient temperature. Also, since water vapor sorbs and desorbs from surfaces, allow plenty of time for the reading to stabilize. Set the flow rate from 0.5 to 1.0 lpm. Check the analyzer reading in the software dashboard.

Note: In general, if reliable calibration standards are not available or if there is not enough time to do the job properly, it is better to leave the zero and span settings alone than to rush through the procedure and make incorrect settings.

Span is a linear function of absorptance, so there is an offset term and a slope term. Both are determined at the factory, and when a (normal) span is set by the user, only the offset term changes. The slope term can be changed by performing a secondary span at a much different concentration than the previous (normal) span.

Step-by-step calibration instructions

To adjust the zero and span calibration:

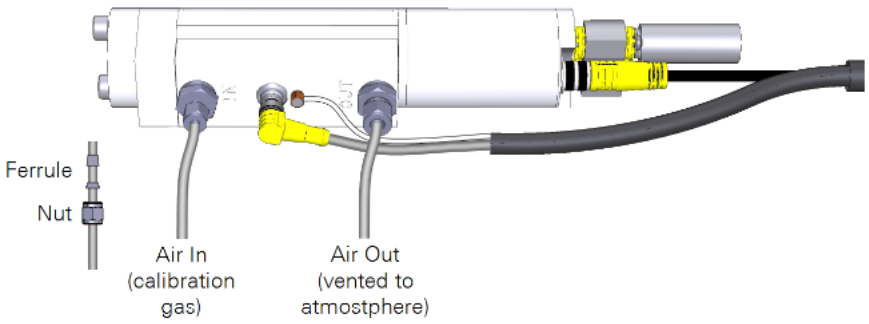


Figure 15: Flow calibration gas into the Air IN port. Be very careful to not damage the inlet thermocouple when connecting the tubing. Attach 15 cm (6 in.) of tube to the outlet to prevent diffusion of ambient air into the analyzer

1. **Clean the optical cell and reassemble it.**
2. **Disconnect the intake tube at the air inlet on the sensor head.**
3. **Connect the calibration gas to the air inlet fitting using 3/8" I.D. Bev-a-line tubing, or with ¼" I.D. Bev-a-line tubing and the 3/8-1/4" adapter. You may need to install compression fittings on the inlet and outlet to connect the tubing.**

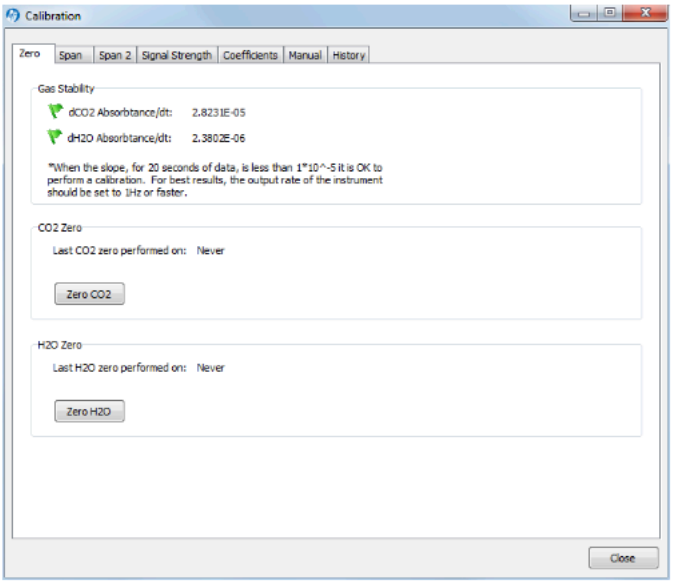
Do not try to insert ¼" Bev-a-line or other small diameter tubing into the air IN or air OUT fittings, as you can break the fine-wire thermocouples in each of the fittings.

4. **Flow CO2-free air at a rate of about 0.5 to 1.0 LPM or more.**
   Attach the zero gas to the air IN fitting. Note the position of the ferrules inside the fitting. When you tighten the fitting, the ferrules will compress slightly to hold the tubing in place.

Zero CO_2

Important: Always zero the instrument before spanning (don't span, then zero).

5. **In the PC software, click *LI-7200/RS > Calibration*.**
   Verify that temperature and pressure sensors are working properly by checking their values in the dashboard.
6. **Click on the *Manual* tab and view the value of Z_{co} (CO_2 Zero).**
7. **Click the *Zero* tab.**
   When the reading has stabilized in the dashboard, and the Gas Stability flag is green, click Zero CO_2 to set the CO_2 zero.



After a brief delay, the displayed CO_2 value should be fluctuating around zero. Check the value of Z_{co} shown on the *Manual* tab. It should be near 1 (typically between 0.85 and 1.1). This value will steadily increase over time (2-3 months) as the internal chemicals lose effectiveness.

Zero H_2O

Now is a good time to check and set the **H_2O** zero, if needed.

8. **Click the *Manual* tab and note the present value of Z_{wo} (H_2O Zero).**
   Wait for the H2O reading to stabilize in the dashboard (3 or 4 minutes). The *Gas Stability* flag will turn from a red 'X' to a green flag when it is OK to perform the calibration.

9. **Click the *Zero* tab and click *Zero H_2O*. Note the new value of Z_{wo} (typically between 0.9 and** 1.2).

Span CO_2

10. **Flow a CO_2 span gas through the optical path at 0.5 to 1 liter/minute.**
11. **Click on the *Span* tab and enter the mole fraction in the 'Span gas concentration' field.**
12. **When stable (1-2 minutes) click *Span CO_2*.**

Click the *Manual* tab and check the new CO_2. Click the *Manual tab* and check the new CO_2 Span value S_c (typically 0.9-1.1).

Span H_2O

13. **To set the H_2O span, flow air of known dew point at about 0.5 to 1.0 LPM.**
    To prevent condensation, use a dew point temperature several degrees below the ambient temperature

14. **Click the Manual tab, and note the present value of the H_2O Span value S_w.**
    Go back to the *Span* tab and enter the span gas dew point temperature in the *Dew point temperature* field.

15. **Observe the H_2O dew point in the dashboard and wait for it to stabilize.**
    The *Gas Stability* flag will turn from a red 'X' to a green flag when it is OK to perform the calibration.

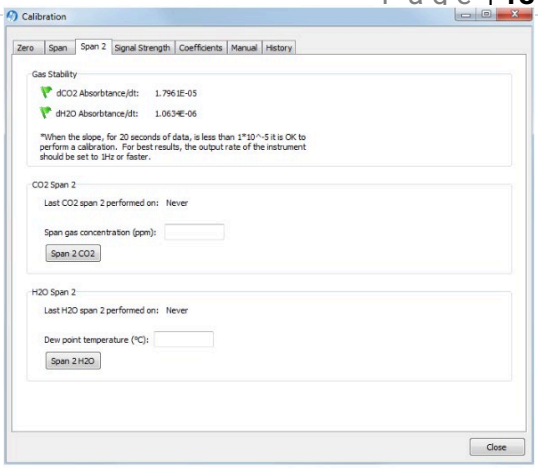
16. **When the reading has stabilized, click *Span* H_2O**
    Click the *Manual* tab again and note the new H_2O Span value S_w (typically 0.9-1.1).

Considerations for setting the secondary span

If you find that after zeroing then spanning at one concentration, the instrument is not within specifications at a different concentration, a secondary span may be in order. The most common reason for doing this would be after a change in the chopper housing temperature set point. The two gas concentrations used for the span and secondary span should be as far apart as possible. For example, at the factory, this is typically done for CO_2 using 200 ppm and 3000 ppm. At a minimum, the pair should be at least 500 ppm apart and bracket your intended operating range (e.g. 300 ppm and 800 ppm). For H_2O, you are constrained by the temperature of the air. You would normally choose a very low dewpoint, such as 5 °C, and something close to (just below) ambient. If the air temperature is 15 °C or less, you should probably avoid doing a secondary water vapor span.

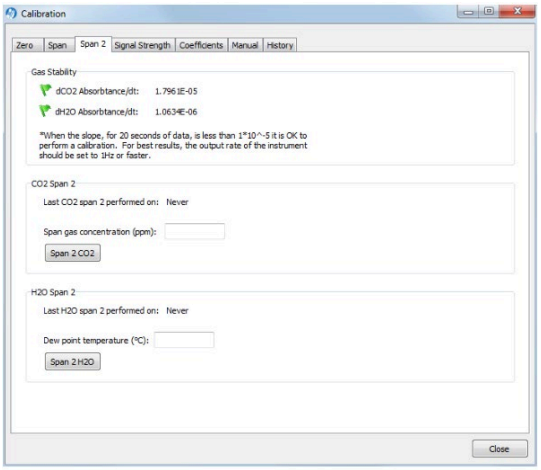
Secondary CO_2 Span

1. **Zero the CO2 reading (see *Zero CO2*)**
2. **Span the CO_2 reading at a concentration below your normal operating range.** (Something in the 200-300 ppm range)
3. **Flow a CO_2 concentration that is at or above your upper operating range** (Above 800 ppm)
4. **Click on the Span 2 tab and when stable, click** *Span 2 H2O*



Secondary H_2O Span

1. **Zero the H_2O reading (see *Zero* H_2O)**
2. **Span the H_2O reading at a dew point that is just below the ambient temperature**
3. **Flow a very low dew point through the tubing**
4. **Click on the *Span 2 tab* and when stable, click *Span 2 H_2O***
   It may take another 10 minutes or so to ensure equilibrium is reached when changing from one concentration to another, due to Water present in the tubing, etc.



Note:

For both the CO2 and H2O secondary spans, you can reverse the concentrations if you wish. The normal span can be the high concentration, and the secondary the low concentration. Also, multiple secondary spans can follow a normal span. Simply make sure that the concentration difference between the normal pan and any secondary span is large.